

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	June - 10 - 2026
Team ID	SWTID-2026-9984
Project Name	BOOK STORE
Maximum Marks	4 Marks

Functional Requirements

The functional requirements of the proposed BookEase online marketplace are categorized as follows:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Accounts	- Registration through forms for Buyers, Sellers, and Admins. - Secure login via email and password. - Edit Profile details (Name, Address).
FR-2	Book Exploration	- Browse complete catalog on Home Page. - Search books by title or author. - Filter books by genre. - Detailed view (title, cover image, stock level, format types, price, reviews).
FR-3	Inventory Management	- Add new books (Cover image upload, title, author, genre, price, stock, formats). - Update existing listings (stock adjustments, details edit). - Delete books from inventory.
FR-4	Cart & Ordering	- Add books to Cart with custom quantity and format (Paperback vs E-Book). - Cart summary checking. - Checkout process capturing delivery address (Flat No, Pincode, City, State). - Order confirmation displaying delivery estimate date.
FR-5	Seller Dashboard	- View order requests matching the seller's catalog. - Modify order fulfillment status (Pending, Shipped, Delivered, Cancelled).
FR-6	Reading Log & Social Interaction	- Rate books (1-5 stars). - Write textual book reviews. - Update progress bar showing

		0% to 100% of reading completed.
FR-7	Wishlist management	- Toggle addition or removal of books to a private wishlist screen.
FR-8	Administration Portal	- Manage lists of registered buyers and sellers. - Vet and approve sellers before they list books. - Delete orders or users.

Non-functional Requirements

The operational constraints and quality attributes of the proposed BookEase system are listed below:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Clean, responsive modern web UI utilizing React, TailwindCSS, Bootstrap, and React-Icons. Layouts adapt automatically to mobile, tablet, and desktop screens.
NFR-2	Security	Cryptographic passwords storage (hashed in the database), CORS protection configured at backend server to only permit verified frontend domains, and endpoint verification.
NFR-3	Reliability	Transaction consistency. Inventory stock is decremented immediately upon successful checkout, preventing double-purchasing of limited stock.
NFR-4	Performance	Sub-second page loads powered by Vite asset compilation and indexing on critical MongoDB fields like `userId`, `bookId`, and `email`.
NFR-5	Availability	Decoupled client-server design. High database availability hosted on MongoDB Atlas cloud nodes.
NFR-6	Scalability	3-Tier MVC architecture separating the frontend client, Node API routers/controllers, and MongoDB storage layer. Allows horizontal scaling of node services.

